

Lending Club Credit Risk Analysis

Loan Pricing vs. Default Risk: A Segmentation Study

Prepared by:	Abdelrahman Mwafy Mechatronics Engineer & Data Analyst
Date:	July 2026
Dataset:	236,846 loans 2014-2016 Source: Lending Club
Tools:	Excel Python (pandas, pyodbc) SQL Server 2022 Power BI Desktop

BUSINESS QUESTION

Are Lending Club borrowers being charged interest rates proportional to their actual default risk - and if not, which segments are most mispriced?

METHODOLOGY

Historical loan data (236,846 records, 27 fields) was loaded into SQL Server 2022 via a Python pipeline using pandas and pyodbc. Five analytical SQL queries produced segmentation results by sub-grade, DTI bucket, verification status, and loan purpose. Findings were visualized in a 4-page Power BI dashboard with DAX measures. **Default rate** = proportion of loans with a defaulted outcome (loan_status = 0). **Pricing gap** = default rate minus interest rate; a positive value indicates underpriced risk.

FINDINGS

1. Grade-level pricing works directionally but underprices high-risk tiers.

Interest rates rise correctly from Grade A (avg 5.51%) to Grade G (avg 27.68%). However, default rates accelerate faster than pricing at D, E, F, and G grades. By Grade G, the average default rate reaches 60.3% against an interest rate of only 26.7% - a 33.6 percentage point gap. Two sub-grade inversions were also identified: D3 charges more than D2 but defaults slightly less (30.5% vs 31.2%), and F3 similarly overcharges relative to F2.

2. Small business loans carry the largest purpose-level pricing gap.

Small business borrowers default at 27.48% but are charged only 16.70% - a gap of **10.78 percentage points**, the largest mismatch across all loan purposes. Car loans are nearly perfectly priced (gap 0.28pp). Debt consolidation - 138,371 loans (58% of the portfolio) - carries a 5.29pp gap, representing significant aggregate exposure. The portfolio-level pricing gap is 4.24pp overall.

Loan Purpose	Loans	Int Rate %	Default %	Gap (pp)
small_business	2,274	16.70	27.48	+10.78
house	878	17.16	23.58	+6.42
debt_consolidation	138,371	12.98	18.27	+5.29
renewable_energy	128	15.91	21.09	+5.18
moving	1,480	15.37	19.73	+4.36
credit_card	56,325	10.98	13.25	+2.27
car	2,152	11.89	12.17	+0.28

Table 1: Pricing gap by loan purpose, ordered by gap size

3. DTI ratio predicts default independently of grade - and is not priced in.

Within Grade D, high DTI (>30%) borrowers default at **39.07%** vs **25.00%** for low DTI (<15%) borrowers - a 14-point spread - despite paying identical average rates (~17%). In Grade C the gap is 22.21% vs 14.10% at the same ~13% rate. DTI is a statistically significant independent risk factor not captured once a grade is assigned.

4. Verification paradox: verified borrowers default more than unverified borrowers.

Across all grade levels, Not Verified borrowers consistently default at lower rates than Verified borrowers. Grade C: Not Verified 15.15% vs Verified 18.25%. Grade D: Not Verified 27.15% vs Verified 31.28%. Pattern holds with robust sample sizes across every grade. Most likely explanation: verification is triggered by existing risk flags, making the Verified group a self-selected higher-risk pool.

RECOMMENDATIONS

- 1. Reprice small business loans upward by 8-10 percentage points**, or implement stricter approval criteria (business plan, minimum revenue verification) to reduce default exposure in this segment.
- 2. Introduce a DTI premium within grades C and D.** Borrowers with DTI above 30% should carry a 2-4pp rate adjustment above the base grade rate.
- 3. Reconsider the verification framework.** Verification outcomes should feed back into grade assignment rather than serving only as an application flag.

LIMITATIONS

This analysis uses historical data from 2014-2016 and may not reflect current conditions. Loan purpose and DTI are self-reported and subject to misrepresentation. Findings describe correlation, not causation - external factors may explain part of the small business default premium beyond pricing model design.